Mathematically, the consumer's surplus (CS) can be defined as

CS = (Area under the demand curve from $x = 0$ to $x = x_0$) – (Area of the rectangle OX_0BP_0)

$$CS = [\int_0^{x0} p(x)\,dx] - x_0p_0$$

Example:12.28

If the demand function is $P = 35 - 2x - x^2$ and the demand x_0 is 3, what will be the consumer's surplus?

Solution

Given demand function,

$$P = 35 - 2x - x^2$$

for $x = 3$

$$P = 35 - 2(3) - 3^2$$

$$= 35 - 6 - 9$$

$$P = 20$$

Therefore,

CS = (Area of the curve below the demand curve from 0 to 3) – Area of the rectangle ($20 \times 3 = 60$)

$$CS = \int_0^3 (35 - 2x - x^2)dx - (20 \times 3)$$

$$= \left[35x - 2\frac{x^2}{2} - \frac{x^3}{3}\right]_0^3 - 60$$

$$= 35(3) - 2(\frac{3^2}{2}) - \frac{3^3}{3} - 60$$

$$= 105 - 9 - 9 - 60$$

$$= 27 \text{ Units.}$$

12.5.6 Producer's surplus

$$PS = P_o x_o - \int_o^{x_o} g(x)\,dx$$

Example 12.29

Given the demand function $P_d = 25 - Q^2$ and the supply function $P_s = 2Q + 1$. Assuming pure competition, find (a) consumers surplus and (b) producers surplus. (P_d = Demand price; P_s = Supply price)

Solution:

For market equilibrium, $P_d = P_s$

$$25 - Q^2 = 2Q + 1$$

$$0 = -25 + Q^2 + 2Q + 1$$

$$0 = -24 + Q^2 + 2Q$$

$$Q^2 + 2Q - 24 = 0$$

$$Q^2 + 6Q - 4Q - 24 = 0$$

$$Q(Q + 6) - 4(Q + 6) = 0$$

$$(Q + 6)(Q - 4) = 0$$

So, $Q = 4$ or $Q = -6$. Since Q cannot be equal to –6,

$$Q = 4$$

When Q=4, $P_d = 25 - 4^2 = 9$;

$$P_s = 2(4) + 1 = 9$$

Consumers' surplus $= \int_0^4 (25 - Q^2)dQ - (9 \times 4)$

$$= \left[25Q - \frac{Q^3}{3}\right]_0^4 - 36$$

$$= [(25)(4) - \frac{1}{3}(4)^3] - (0) - 36$$

$$= [100 - \frac{64}{3}] - (0) - 36 = 42.67$$

Producers' surplus PS

$$(PS) = (9 \times 4) - \int_0^4 (2Q + 1)dQ$$

$$= 36 - \left[(Q^2 + Q)\right]_0^4$$

$$= 36 - (16 + 4) = 16$$

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